A cubic equation with no integer solution
Ireland–Rosen, 3.5

Theorem 1. *The equation $7x^3 + 2 = y^3$ has no solution in integers.*

Proof. Suppose, for a contradiction, that $x, y \in \mathbb{Z}$ satisfy

$$7x^3 + 2 = y^3.$$

We first observe that no cube in $\mathbb{Z}/7\mathbb{Z}$ is equal to 2. Indeed, every element of $\mathbb{Z}/7\mathbb{Z}$ is represented by one of $0, 1, 2, 3, 4, 5, 6$, and the cubes of these representatives modulo 7 are

z	0	1	2	3	4	5	6
z^3	0	1	1	6	1	6	6

Thus for every $z \in \mathbb{Z}/7\mathbb{Z}$ we have $z^3 \neq 2$; in fact, the only cubes modulo 7 are 0, 1, and 6.

Now reduce the assumed equation modulo 7. In $\mathbb{Z}/7\mathbb{Z}$, we have $7 = 0$, so the image of the left-hand side $7x^3 + 2$ is

$$7x^3 + 2 = 0 \cdot x^3 + 2 = 2.$$

Hence the image of y^3 in $\mathbb{Z}/7\mathbb{Z}$ is equal to 2, i.e.,

$$y^3 = 2 \qquad \text{in } \mathbb{Z}/7\mathbb{Z}.$$

This contradicts the observation above that no cube in $\mathbb{Z}/7\mathbb{Z}$ equals 2. Therefore no integers x and y can satisfy $7x^3 + 2 = y^3$. □